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AREAS OF RESEARCH

Machine Learning; Deep Learning; Bayesian Modeling; Survey Sampling Methodology; Social Network Analysis; Randomized Controlled Trials.

ACADEMIC APPOINTMENTS

2025–present	**Assistant Professor** Department of Mathematics and Statistics, North Carolina Agricultural and Technical State University
2025–present	**Affiliate Faculty** Center on Human Trafficking Research and Outreach, University of Georgia
2020–2025	**Data Scientist** School of Social Work, University of Georgia
2019–2020	**Visiting Research Scholar** Department of Statistics, Virginia Tech
2015–2020	**Assistant Professor (tenured)** International Transportation Engineering, School of Highway, Chang'an University, China

EDUCATION & TRAINING

2014	**Ph.D.,** Genetics, Bioinformatics and Computational Biology with Concentration in **Statistics** Virginia Tech *Committee:* Prof Ina Hoeschele, Prof Xinwei Deng, Dr. Hongxiao Zhu *Dissertation*: Assessment of Sparse Regression Machine Learning Methods for Genome-Wide Association Studies
2011	**M.S., Statistics** Virginia Tech
2009	**B.S., Applied Mathematics** Beijing Normal University, China
2026	**Faculty Cohort, Amazon Web Services Machine Learning University** Generative AI and Agentic AI workshops

FELLOWSHIPS & AWARDS

2024–2025	**Faculty Fellow**, Institute of Artificial Intelligence, University of Georgia Role: Research Faculty

2024	**Team Impact Award**, Center on Human Trafficking Research & Outreach, University of Georgia Role: Co-Investigator
2021–2025	**Faculty Fellow**, Owens Institute for Behavioral Research, University of Georgia Role: Research Faculty
2017	**Excellent Supervisor Award**, 11th Challenge Cup College Students Science and Technology Work Competition of Shaanxi Province, Xi'an, China Ji, Q. et al. Sponge city with starlight road: High brightness asphalt concrete pavement based on waste glass recycling (2nd Prize). Role: Co-Supervisor
2009–2010	**Graduate Student Fellowship**, Virginia Tech ($25,000)
2007	**Championship**, Contemporary Undergraduate Mathematical Contest in Modeling, Beijing, China Role: Team Leader
2005–2009	**Best Bachelor's Thesis of the Year Award** **Outstanding Undergraduate Research Award** **Outstanding Academic Contest Team Award** **Excellent Summer Internship Group** Beijing Normal University, China
2005	**Scholarship for Outstanding Students in the National College Entrance Exam**, China Construction Bank Top 0.2% of students nationally

FUNDED RESEARCH PROJECTS

Currently Co-Principal Investigator on a $399,999 National Science Foundation project. Between 2020 and 2025, served as Co-Investigator on a $7.7 million project and as Quantitative Research Scientist on U.S. Department of State–funded projects totaling more than $24 million at the University of Georgia. A Principal Investigator on $15,000 in funded projects at Chang'an University, China. During graduate studies at Virginia Tech, supported National Institutes of Health–funded research projects totaling $1.78 million.

External Grants

1. 2025–2028 **Co-PI**
   Team: Abeer Hasan (PI), Sayed Mostafa (Co-PI), Tamer Elbayoumi (Co-PI)
   Source: National Science Foundation, Division of Undergraduate Education, Grant #2510214
   Title: Targeted Infusion Project: Infusing Responsible Generative AI Education into Undergraduate Statistics and Data Science Curricula (AI4DS)
   Amount: $399,999
   Description: Enhances undergraduate data science education by integrating generative AI into core courses: promoting AI literacy and ethical awareness, equipping students with practical skills to integrate AI tools in data science careers, and training faculty and graduate teaching assistants in effective practices for leveraging AI in STEM education.

2. 2024–2025 **Co-I**
Team: David Okech (PI), Anna Cody (Co-I)
Source: U.S. Department of State, Office to Monitor and Combat Trafficking in Persons, Grant #SSJTIP22CA0029
Title: Reducing the Prevalence of Human Trafficking through Increasing the Financial Literacy and Capability of Survivors and Those At-Risk in Sub-Saharan Africa
Amount: $5,200,000
Description: Financial capability interventions for labor trafficking survivors and those at risk in selected districts of Zambia and Malawi, evaluated with a stepped-wedge cluster randomized controlled trial comparing two stacked intervention programs, to reduce domestic and cross-border labor trafficking among youth and young adults.

3. 2022–2024 **Co-I**
Team: David Okech (PI), Lydia Aletraris (Co-I), Anna Cody (Co-I), Laura Zimmermann (Co-I)
Source: U.S. Department of State, Office to Monitor and Combat Trafficking in Persons, Grant #SSJTIP22CA0029
Title: Reducing the Prevalence of Human Trafficking through Increasing the Financial Literacy and Capability of Survivors and Those At-Risk in Sub-Saharan Africa (Phase I)
Amount: $2,200,065
Description: Mixed-methods study using a household survey with equal probability of selection and respondent-driven sampling to estimate the prevalence of labor trafficking among youth and young adults (aged 16–37) in Zambia and Malawi who returned from work in South Africa, as well as those at risk.

4. 2019–2025 **Quantitative Research Scientist**
Team: David Okech (PI), Jody Clay-Warner (Co-I), Alex Balch (Co-I, University of Liverpool), Lydia Aletraris (Co-I), Tamora Callands (Co-I)
Source: U.S. Department of State, Office to Monitor and Combat Trafficking in Persons, Grant #SSJTIP19CA0032
Title: Measuring the Prevalence of Trafficking around the World: Implications for Research, Programming, and Policy
Amount: $15,750,000 + $4,000,000 (awarded in 2020)
Description: Scales up research and programming in Sierra Leone and Guinea and starts new work in Senegal: establishing robust baseline prevalence data in hot spots, enhancing implementing partners' anti-slavery operations, and documenting the strengths and weaknesses of prevalence estimation methods for hard-to-reach populations through a Prevalence Reduction Innovation Forum.

5. 2018–2024 **Quantitative Research Scientist**
Team: David Okech (PI), Jody Clay-Warner (Co-PI), Alex Balch (Co-PI, University of Liverpool), Tamora Callands (Co-I), Nate Hansen (Co-I)
Source: U.S. Department of State, Office to Monitor and Combat Trafficking in Persons, Grant #SSJTIP18CA0015
Title: Anti-trafficking Research and Programming in West Africa
Amount: $2,200,065

Description: Establishes robust baseline data on the prevalence of child trafficking in selected hot spots in Sierra Leone and Guinea and enhances implementing partners' anti-slavery operations.

6. 2019–2020 **Co-I**
   Team: Qing Zhang (PI)
   Source: Transportation Construction Group Corporation of Shaanxi Province and Chang'an University, Ministry of Transport, China
   Title: Writing for the National Standard of Highway Construction Project Management
   Amount: $140,000
   Description: Wrote a national standard of highway construction project management applicable across construction contexts and settings for the next five years.

7. 2017–2019 **PI**
   Source: Department of Science and Technology of Shaanxi Province, China, Grant #2017JQ5117
   Title: Deep Correlation and Information Mining of Massive Multi-source Road Performance Data
   Amount: $5,000
   Description: Mined multi-source road performance data and optimized maintenance and rehabilitation plans over years based on temporal-spatial models.

8. 2016–2017 **PI**
   Source: Key Laboratory Project of Hebei Province of Shijiazhuang Tiedao University and Chang'an University, China, Grant #JTKY2016002
   Title: Study on Estimation Methods of Missing Data in Road Performance Monitoring
   Amount: $3,000
   Description: Analyzed dynamic real-time data in road performance monitoring and diagnostics; developed and systematically evaluated statistical models for imputation of missing data.

9. 2015–2017 **PI**
   Source: Basic Scientific Research of Central Universities, Ministry of Education of China, Grant #310821161029
   Title: Dynamic Reliability Assessment of Road Infrastructure using a Bayesian Time-spatial Method
   Amount: $7,500
   Description: Examined the reliability and accuracy of the road infrastructure database and improved its usability by employing a time-spatial model to impute missingness.

10. 2010–2014 **Graduate Research Assistant**
    Team: Ina Hoeschele (PI)
    Source: National Institutes of Health, Grant #1R01HG005254-01
    Title: Highly Multivariate Quantitative Trait Loci Mapping in Systems Genetics
    Amount: $773,732
    Description: Simulated benchmark datasets in gene network inference and verified methods for systems genetics data.

11. 2011–2014 **Graduate Research Assistant**
Team: Yongmei Liu (PI), Ina Hoeschele (Co-PI)
Source: National Institutes of Health, Grant #5R01AG032098
Title: Genetic Determinants of Visceral Adiposity: A Genome-Wide Association Study
Amount: $449,060
Description: Established robust and effective penalized regression methods for genome-wide association studies, from which genetic determinants of visceral adiposity were identified.

12. 2009–2014 **Graduate Research Assistant**
Team: Yongmei Liu & Ina Hoeschele (PIs)
Source: National Institutes of Health, Grant #1R01HL101250-01
Title: Epigenome-wide Association Study of DNA Methylation and Atherosclerosis
Amount: $559,641
Description: Established the epigenome-wide association of DNA methylation and atherosclerosis; identified epigenetic biomarkers that informed follow-up verification experiments.

Internal Grants

13. 2025–2027 **PI**
Source: North Carolina A&T State University, Faculty Startup Grant
Title: Scalable Machine Learning and Adaptive Network Sampling Analysis for Hidden and Hard-to-Reach Populations
Amount: $109,333
Description: A multi-stream deep learning approach to social network analysis integrating Exponential Random Graph Models, Reinforcement Learning, and Graph Neural Networks. Using network scale-up and respondent-driven sampling data collected in West and Southern Africa, develops robust methods for accurate prevalence estimates in resource-poor and data-deficient environments to guide policy and decision-making.

14. 2024–2025 **Team Member**
Team: Orion Mowbray (Team Leader)
Source: University of Georgia, Office of Research, Teaming for Interdisciplinary Research Pre-Seed Program
Title: Data Science and AI for Social Impact (https://research.uga.edu/team-pre-seeds/projects/data-science-and-ai-for-social-impact/)
Amount: $4,500

Grant Proposals Under Review

1. To submit October 2026 **PI**
Team: Carmen Monico (Co-PI), Jenny Afkinich (Co-PI), Meredith Peffley (Subaward-PI)
Source: National Institute of Justice, Office for Victims of Crime, FY26 Preventing Trafficking of Girls
Title: PROTECT Girls-NC: Strengthening Family, School, and Community Protective Systems to Prevent Sex Trafficking of Girls in North Carolina
Amount: $500,000

Description: The project prevents sex trafficking of girls in North Carolina in three ways: it equips at-risk girls and caregivers to recognize and resist exploitation; it empowers a new Survivor Advisory Group (an adult survivor of child sex trafficking, a survivor's caregiver, and the Survivor Advocate), the Survivor Outreach Educator, and school staff as lasting community prevention leaders; and it extends this network into two Guilford County public schools.

2. Submitted Aug 2026 **Co-PI**
   Team: Sayed Mostafa (PI), Guoqing Tang (Co-PI), John P. Roop (Co-PI), Liu Liu (Subaward-PI)
   Source: U.S. National Science Foundation
   Title: Computational Experiences in Introductory Mathematics: A Longitudinal Study of Mathematical Agency and STEM Progression
   Amount: $1,500,000
   Description: A sixty-month ECR project will develop and establish validity evidence for a common multidimensional framework for characterizing undergraduate student computing-empowered math learning experiences and examine how their intensity, accumulation, timing, and continuity are associated with mathematical agency and related mechanisms, mathematical learning, and STEM progression.
3. Submitted Aug 2026 **Co-PI**
   Team: Sayed Mostafa (PI), Guoqing Tang (Co-PI), John P. Roop (Co-PI), Barbara Tankersley (Co-PI)
   Source: U.S. Department of Education
   Title: AI-Enhanced Mathematics for STEM Success: Integrating Machine Learning, Mathematical Modeling, and Data Science in Undergraduate Education and Research
   Amount: $600,000
   Description: A twelve-month initiative integrating AI-enhanced computational and data-driven approaches into gateway mathematics instruction, faculty and graduate student development, pre-college enrichment, and undergraduate research.
4. Submitted Aug 2026 **Co-PI**
   Team: Sayed Mostafa (PI), Guoqing Tang (Co-PI), John P. Roop (Co-PI), Barbara Tankersley (Co-PI)
   Source: U.S. Department of Education
   Title: AI-Enhanced Mathematics for STEM Success: Integrating Machine Learning, Mathematical Modeling, and Data Science in Undergraduate Education and Research
   Amount: $600,000
   Description: A twelve-month initiative integrating AI-enhanced computational and data-driven approaches into gateway mathematics instruction, faculty and graduate student development, pre-college enrichment, and undergraduate research.
5. Submitted June 2026 **PI**
   Team: Carmen Monico (Co-PI), Jenny Afkinich (Co-PI)
   Source: National Institute of Justice, FY25 Research and Evaluation on Human Trafficking
   Title: A Multilevel System Evaluation of Coordinated, Wraparound Service Access, Delivery, Implementation, and Outcomes for Survivors of Sex Trafficking in the U.S. South
   Amount: $850,704
   Description: Assesses, intervenes in, and strengthens the coordination and provision of wraparound services across agencies for survivors and those at risk of sex trafficking in

North Carolina, in collaboration with the North Carolina Human Trafficking Commission and CrossRoads.

6. Submitted Feb 2026 **Co-PI**
Team: Meng Zhao (PI), Shaohu Zhang (Co-PI)
Source: National Institutes of Health, National Institute on Aging, R21
Title: Development and Pilot Testing: A Home-Based Mixed Reality Square-Stepping Exercise in Older Adults at Risk for Alzheimer's Disease
Amount: $398,750
Description: Tests the feasibility and usability of a Mixed Reality Square-Stepping Exercise program in older adults with self-reported cognitive decline and evaluates its effectiveness on cognitive health and quality of life in a 20-participant pilot with small-sample statistical analyses.

Grant Proposals in Progress

7. Completed, to submit 2027 **PI**
Team: Carmen Monico (Co-PI), Jenny Afkinich (Co-PI)
Source: National Institute of Justice, FY26 Research and Evaluation on Violence Against Women
Title: Expanding Caregiver Capacity and Community Empowerment for Minor Survivors of Sex Trafficking in North Carolina
Amount: $900,000

8. Completed, to submit **PI**
Team: Carmen Monico (Co-PI), Jenny Afkinich (Co-PI)
Source: National Institute of Justice, FY26 Research and Evaluation on Violence Against Women
Source: National Institute of Justice, FY26 Research and Evaluation on Violence Against Women
Amount: $1,000,000

9. 2027 **PI**
Source: National Science Foundation, CAREER
Title: CAREER: AI-Assisted Adaptive Network Sampling for Hidden and Hard-to-Reach Populations
Amount: $650,000

10. 2027 **PI**
Team: Xueying Liu (Co-PI)
Source: National Science Foundation 23-598, Historically Black Colleges and Universities Excellence in Research
Title: Reinforcement Learning-Deep Learning for Social Network Analysis
Amount: $650,000

11. 2027 **Co-PI**
Team: Charnell Long (PI)
Source: National Science Foundation 26-501, Collaboratory to Advance Mathematics Education and Learning (CAMEL) for K-12
Title: CAMEL-CN: Constructing a Dataset to Explore Learning Trajectories in Southern Mathematics Education
Amount: $750,000

PUBLICATIONS

Peer-Reviewed Journal Articles

1. **Yi, H.**, Li, J., Datta, G., Okech, D., & Clay-Warner, J. (2026). A novel application of small area estimation methodology to study child-trafficked population in Sierra Leone. *Journal of Computational Social Science*. https://doi.org/10.1007/s42001-026-00476-1 (IF: 2.30)
2. **Yi, H.**, Liu, X., Deng, X., Liu, L., & Okech, D. (2026). New perspectives of respondent-driven sampling with application to a sex-trafficked population in Senegal. *Journal of Statistical Theory and Practice*. https://doi.org/10.1007/s42519-026-00612-0 (IF: 0.90)
3. Liu, L., Liu, J., **Yi, H.**, & Okech, D. (2026). Simulation-cum-ROC fit evaluation in sparse ordinal CFA under category collapsing. *Structural Equation Modeling: A Multidisciplinary Journal*. https://osf.io/b7atw/overview (IF: 3.20)
4. Datta, G., Li, J., **Yi, H.**, & Okech, D. (2025). Small area estimation of child trafficking proportions in chiefdoms of Sierra Leone and overall ranking of the chiefdoms. *Journal of Survey Statistics and Methodology*. https://doi.org/10.1093/jssam/smaf013 (IF: 2.45)
5. Clay-Warner, J., **Yi, H.**, Li, J., Kawashima, T., Okech, D., & Konteh, F. H. (2025). A comparison of top-coding strategies in scale-up estimation of personal network size. *Social Networks*. https://www.sciencedirect.com/science/article/abs/pii/S0378873325000371 (IF: 2.38)
6. Schroeder, E., **Yi, H.**, Okech, D., Bolton, C., Aletraris, L., & Cody, A. (2024). Do social service interventions for human trafficking survivors work? A systematic review and meta-analysis. *Trauma, Violence, & Abuse, 25*(3), 2012–2027. https://doi.org/10.1177/15248380231204885 (IF: 7.00)
7. Cody, A. M., Okech, D., **Yi, H.**, Aletraris, L., Clay-Warner, J., & Callands, T. (2024). When child trafficking and informal fostering intersect: A mixed methods study. *Social Work and Social Sciences Review, 24*(2), 33–61. https://doi.org/10.1921/swssr.v24i2.2026 (IF: 0.35)
8. **Yi, H.**, Vincent, K., Okech, D., Clay-Warner, J., Li, J., Kawashima, T., … & Hassan Konteh, F. (2023). An empirical comparison of a traditional strategy and network scale-up method for prevalence estimation of child trafficking in Sierra Leone. *Crime & Delinquency*. https://doi.org/10.1177/00111287231170122 (IF: 2.85)
9. Li, D., Ranjitkar, P., Zhao, Y., **Yi, H.**, & Rashidi, S. (2017). Analyzing pedestrian crash injury severity under different weather conditions. *Traffic Injury Prevention, 18*(4), 427–430. https://doi.org/10.1080/15389588.2016.1207762 (IF: 1.41)

10. **Yi, H.**, Breheny, P., Imam, N., Liu, Y., & Hoeschele, I. (2015). Penalized multimarker vs. single-marker regression methods for genome-wide association studies of quantitative traits. *Genetics, 199*(1), 205–222. https://doi.org/10.1534/genetics.114.167817 (IF: 4.40)
11. Guo, C., Xu, L., Yang, S., **Yi, H.**, & Zheng, X. (2009). Analysis and research on the extreme value of multivariable functions (in Chinese). *Sciencepaper Online, 2*(1), 15–24. ISSN: 1673-7180. https://highlights.paper.edu.cn/index/paper_detail/3120 (IF: 0.87)
12. Wang, Q., **Yi, H.**, Liu, L., & Meng, D. (2008). Progress in gene logic networks (in Chinese). *Progress in Biochemistry and Biophysics, 35*(11), 1239–1246. https://www.pibb.ac.cn/pibbcn/article/abstract/20080155 (IF: 0.20)

Manuscripts Under Review

13. Liu, L., & **Yi, H.** Evaluating covariance priors for inference on random-slope heterogeneity in Bayesian multilevel models. *Multivariate Behavioral Research* (submitted October 2026)
14. Okech, D., **Yi, H.**, Goulart, P., & Konteh, F. H. Vulnerability of children with disabilities in trafficking: A low-resourced country study. *Children and Youth Services Review* (IF: 3.30; submitted October 10, 2025). https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=5701717
15. Liu, L., Liu, J., **Yi, H.**, & Yang, P. Analyzing pretest-posttest data: A design-first, estimand-first tutorial. *The Quantitative Methods for Psychology* (IF: 1.30; submitted April 12, 2026).

Manuscripts in Preparation

16. **Yi, H.**, Liu, L., Okech, D., & Olalere, A. An explainable artificial intelligence framework for child trafficking risk assessment.
17. Liu, L., **Yi, H.**, & Okech, D. Interpretable variance-prior calibration for Bayesian multilevel logistic regression models.
18. **Yi, H.**, Liu, X., & Deng, X. Elastic respondent-driven sampling method.
19. Liu, X., **Yi, H.**, Deng, X., & Okech, D. Respondent-driven sampling versus link-tracing sampling: Exploring seeding effects and length of referral-chain to prevalence estimation.
20. **Yi, H.**, Okech, D., Liu, X., & Liu, L. Improving prevalence estimation for hidden populations: An empirical evaluation of respondent-driven and link-tracing sampling strategies.
21. **Yi, H.** AI and survey methods for hard-to-reach populations.
22. **Yi, H.** Estimating the prevalence of human trafficking in West Africa: A comparative analysis of respondent-driven sampling and network scale-up frameworks.
23. **Yi, H.** Tackling data sparsity in hidden population research: Methodological lessons from the field.
24. **Yi, H.** The role of homophily in respondent-driven sampling: Implications for victim identification.

Book Chapters

Yi, H., & Liu, N. (2018). Chapter 5. Writing for project management. In G. Yang & C. Li (Eds.), *Applied English writing in civil engineering* (in English). People's Communications Publishing House. ISBN: 9787114147395.

Dissertation

Yi, H. (2014). *Assessment of penalized regression for genome-wide association studies* [Doctoral dissertation, Virginia Tech]. https://vtechworks.lib.vt.edu/server/api/core/bitstreams/d5549f66-a14e-467f-93ad-390eaf09b227/content

Conference Proceedings

Guan, J., **Yi, H.**, Zhang, Q., & Zhang, Z. (2020). A spatial-temporal optimization model for road network maintenance and rehabilitation decision-making. *Proceedings of the 20th COTA International Conference of Transportation Professionals (CICTP 2020)*.

Policy Briefs & Technical Reports

1. Okech, D., Cody, A., Chunga, J., Zimmermann, L., Tsoka, M., Alfonso, W., Thakwalakwa, C., **Yi, H.**, Makonyola, A., Waswa, A., Goulart, P., Ager, M., Bolton, C., Dockweiler, H., Aletraris, L., Eng, C. L., Schroeder, E., Gordon, M. E., Ansong, D., Okumu, M., Johnson, L., Appiah-Kubi, J., Muzei, J., Koomson, I., Sherraden, M. S., & Sherraden, M. (2025). *Labor trafficking in Malawi: Prevalence, vulnerabilities, and recommendations for financial capability intervention among youth and young adults in Malawi, a baseline study* [107 pages]. Center on Human Trafficking Research and Outreach, University of Georgia. https://doi.org/10.71927/uga.27585
2. Okech, D., Cody, A., Chibwili, E., Zimmermann, L., Bwalya, R., Simbaya, J., **Yi, H.**, Mulambia, C. N., Waswa, A., Goulart, P., Ager, M., Bolton, C., Pengele, M., Dockweiler, H., Aletraris, L., Eng, C. L., Schroeder, E., Gordon, M. E., Ansong, D., Okumu, M., Johnson, L., Appiah-Kubi, J., Muzei, J., Koomson, I., Sherraden, M. S., & Sherraden, M. (2025). *Labor trafficking in Zambia: Prevalence, vulnerabilities, and recommendations for financial capability intervention among youth and young adults in Zambia, a baseline study* [150 pages]. Center on Human Trafficking Research and Outreach, University of Georgia. https://doi.org/10.71927/uga.27584
3. Okech, D., Cody, A. M., **Yi, H.**, Geibel, S., Gaye, I., Sabe, Z., Ahmed, M., Tun, W., Clay-Warner, J., Callands, T. A., Balch, A., Akomolafe, T., Mbow, F. B., Ball, T. S., Waswa, A. N., Schroeder, E. L., Goulart, P., Eng, C. L., Manga, G. V., Bolton, C., Onyioha-Clayton, N., & Barre, A. D. (2025). *Estimating changes in sex trafficking prevalence, perceptions and policies in gold mining communities in Kédougou, Senegal since 2021: An endline research study* [68 pages]. Center on Human Trafficking Research & Outreach, University of Georgia. https://openscholar.uga.edu/record/26742?v=pdf
4. Okech, D., Cody, A. M., Clay-Warner, J., Balch, A., **Yi, H.**, Callands, T., … & Fofanah, U. (2025). *Assessing changes in child trafficking and the worst forms of child labor in Sierra Leone from 2019–2024* [93 pages]. Center on Human Trafficking Research & Outreach, University of Georgia. https://openscholar.uga.edu/record/27593/files/SL%20Endline%20FINAL_11.13-2025_Final%202_DOI.pdf
5. Okech, D., Clay-Warner, J., Balch, A., Callands, T., **Yi, H.**, Cody, A., Bryant, H., & Bolton, C. (2023). *Child trafficking and the worst forms of child labor in Kambia, Sierra Leone: A mixed methods study* [57 pages]. Center for Human Trafficking Research & Outreach,

University of Georgia. https://cenhtro.uga.edu/wp-content/uploads/2025/05/Kambia_Baseline_APRIES.pdf

6. Okech, D., Clay-Warner, J., Balch, A., Callands, T. A., **Yi, H.**, Cody, A. M., Bryant, H., Bangoura, A., Bolton, C., & Hansen, N. (2022). *Child trafficking and child labor in Guinea: A mixed-methods study* [105 pages]. Center for Human Trafficking Research & Outreach, University of Georgia. https://cenhtro.uga.edu/wp-content/uploads/2025/05/Guinea-Baseline-.pdf
7. Okech, D., Kelly, J., Mame Soukeye, M., Cronberg, A., **Yi, H.**, Cody, A., Clay-Warner, J., Woubedle, A., Nupur, K., Shiyun, M., Cheikh Moussa, N., Alioune Badara, T., Balch, A., Bolton, C., Onyioha-Clayton, N., & Diagne Barre, A. (2022). *Sex trafficking in the gold mining areas of Kédougou, Senegal* [88 pages]. Center for Human Trafficking Research & Outreach, University of Georgia. https://cenhtro.uga.edu/wp-content/uploads/2025/05/Senegal_Baseline_Final_24.pdf
8. Cody, A., **Yi, H.**, Okech, D., Balch, A., Barre, A., & Clay-Warner, J. (2022). *Severe forms of sex trafficking in Kédougou, Senegal: Baseline findings summary* [3 pages]. Center on Human Trafficking Research & Outreach, University of Georgia, & Mantle. https://cenhtro.uga.edu/wp-content/uploads/2025/05/Senegal_Baseline_Summary.pdf
9. Bryant, H., Balch, A., Cody, A., & **Yi, H.** (2022). *New research on child trafficking and child labor in Kailahun, Kenema, and Kono: Recommendations for policy* [8 pages]. Center on Human Trafficking Research & Outreach, University of Georgia. https://cenhtro.uga.edu/wp-content/uploads/2025/05/APRIES-SL-Policy-Brief.pdf
10. Okech, D., Clay-Warner, J., Balch, A., Callands, T., **Yi, H.**, Cody, A., & Bolton, C. (2021). *Child trafficking and the worst forms of child labor in Sierra Leone: A mixed methods study* [194 pages]. Center for Human Trafficking Research & Outreach, University of Georgia. https://doi.org/10.71927/uga.26738

PRESENTATIONS

Selected Peer-Reviewed Conference Presentations

1. Liu, L., **Yi, H.**, & Okech, D. (2027, April 14–18). *Interpretable variance-prior calibration for Bayesian multilevel logistic regression models.* Paper submitted to the 2027 American Educational Research Association Annual Meeting, Toronto, Canada. (Draft paper submitted July 21, 2026)
2. **Yi, H.**, Hasan, A., Liu, L., & Olalere, A. (2026, October 6–8). *An explainable artificial intelligence framework for child trafficking risk assessment.* Paper presented at the Fall Technical Conference, Charlotte, NC.
3. **Yi, H.**, Liu, X., & Deng, X. (2026, October 6–8). *Prevalence estimation for sex trafficking in Senegal using respondent-driven sampling.* Paper presented at the Fall Technical Conference, Charlotte, NC.
4. **Yi, H.**, Liu, X., Deng, X., Liu, L., & Okech, D. (2026, August 3–8). *New respondent-driven sampling perspectives for rare and hidden populations* [Poster]. Joint Statistical Meetings, Boston, MA.

5. Okech, D., Liu, L., Cody, A. M., Bolton, C., **Yi, H.**, Waswa, A., & Goulart, P. (2026, June). *Measuring the unseen: Mixed methods and Bayesian multilevel modelling in survivor-centered child trafficking research* [Workshop]. The Joint Conference on Social Work, Education and Social Development (SWSD), Nairobi, Kenya.
6. Okech, D., Chibwili, E., Chunga, J., Cody, A. M., Bolton, C., **Yi, H.**, Liu, L., Waswa, A., & Goulart, P. (2026, June). *Economic migration and labor trafficking: A mixed-methods study exploring intersecting vulnerabilities among youth and young adults in Malawi and Zambia.* Paper presented at the Joint Conference on Social Work, Education and Social Development, Nairobi, Kenya.
7. Liu, L., Cody, A., Okech, D., **Yi, H.**, Waswa, A., & Goulart, P. (2026). *Beyond rescue: Child trafficking, educational access and reintegration in Eastern Sierra Leone.* Paper presented at the American Educational Research Association Annual Meeting.
8. Okech, D., Liu, L., & **Yi, H.** (2026, January 14–18). *Applied Bayesian multilevel modeling in social work: A practical introduction through a child trafficking case study* [Workshop proposal]. Society for Social Work and Research, Washington, DC. (Submitted April 15, 2025; not accepted)
9. Cody, A., Okech, D., & **Yi, H.** (2026, January 14–18). *Constraints on financial well-being and potential for labor trafficking risk among Malawian youth and young adults* [Flash talk]. Society for Social Work and Research, Washington, DC.
10. Okech, D., **Yi, H.**, & Cody, A. (2026, January 14–18). *Growing sex trafficking in Senegal's gold mining areas?: Link-tracing sampling method versus respondent-driven sampling method* [Workshop proposal]. Society for Social Work and Research, Washington, DC. (Not accepted)
11. Okech, D., **Yi, H.**, Cody, A., & Liu, L. (2025, August 21–22). *Between and beyond respondent-driven sampling and link-tracing sampling: Adaptive sampling design and prevalence estimation.* Paper presented at the 2nd International Conference on Research on Hard-to-Reach Populations, Nuffield College, Oxford, UK.
12. **Yi, H.**, Okech, D., Liu, L., Clay-Warner, J., Waswa, A., & Goulart, P. (2025, August 3). *Prevalence estimation for non-probabilistic samples: Link-tracing sampling and respondent-driven sampling.* Paper presented at the Joint Statistical Meetings, Nashville, TN.
13. Liu, L., Okech, D., & **Yi, H.** (2025, August 6). *Informative variance priors for Bayesian multilevel modeling with large and small samples.* Paper presented at the Joint Statistical Meetings, Nashville, TN.
14. **Yi, H.**, Okech, D., Vincent, K., & Clayton, N. (2025, January 16). *Recent advances on estimating population size with link-tracing sampling* [Workshop]. Society for Social Work and Research, Seattle, WA. https://sswr.confex.com/sswr/2025/webprogram/Session15559.html
15. Okech, D., Cody, A. M., **Yi, H.**, Aletraris, L., Bolton, C., Chibwili, E., Chunga, J., Eng, C. L., Goulart, P., & Gordon, M. E. (2025, January 17). *Labor trafficking among youth and young adults in Malawi and Zambia: Understanding the scope and context.* Paper presented at the Society for Social Work and Research, Seattle, WA. https://sswr.confex.com/sswr/2025/webprogram/Paper58234.html

16. Zimmermann, L., Okech, D., **Yi, H.**, & Aletraris, L. (2024, November 2–3). *How widespread is labor trafficking? Evidence from three measurement approaches in two countries.* Paper presented at the North East Universities Development Consortium (NEUDC), Boston, MA.
17. **Yi, H.**, Okech, D., & Clay-Warner, J. (2024, November 7–10). *Data-driven approaches for estimating prevalence of hard-to-reach populations.* The Thirteenth National Conference on Statistics in China, Xiamen, Fujian, China. (Accepted but not presented)
18. **Yi, H.**, Vincent, K., Okech, D., Clay-Warner, J., Onyioha-Clayton, N., & Waswa, A. (2024, August 5). *A novel estimate for the respondent-driven sampling and network sampling methods: A resampling approach.* Paper presented at the Joint Statistical Meetings, Portland, OR.
19. **Yi, H.**, Okech, D., Datta, G., Goulart, P., Li, J., Clay-Warner, J., & Cody, A. (2024, August 7). *A novel application of small area estimation methodology to study child-trafficked population in Sierra Leone.* Paper presented at the Joint Statistical Meetings, Portland, OR.
20. Clay-Warner, J., **Yi, H.**, Li, J., Kawashima, T., Okech, D., & Konteh, F. H. (2024, June 24–30). *"How many people do you know?": Topcoding strategies in the estimation of personal network size.* Paper presented at the International Network for Social Network Analysis, Edinburgh, Scotland.
21. **Yi, H.**, Waswa, A., Eng, C., Cody, A., Goulart, P., & Okech, D. (2024, June 20–21). *"How can emerging data-driven technologies improve the global response to human trafficking?": New advancements in the prevalence estimation in West Africa.* Human Trafficking Research Network Conference, Faculty of Law, University of A Coruña, Spain. (Accepted but not presented)
22. Cody, A., **Yi, H.**, Callands, T., Okech, D., Eng, C., & Gordon, M. (2024, May 28–30). *Moving beyond "informing": Considering the role of qualitative research in mixed methods, applied social science research* [Virtual presentation]. The 20th International Congress of Qualitative Inquiry, Urbana, IL.
23. **Yi, H.**, Aletraris, L., Cody, A., Okech, D., Onyioha-Clayton, N., Goulart, P., & Waswa, A. (2024, April 11–12). *Unveiling the prevalence and drivers of sex trafficking among women in Senegal's gold mining areas: A mixed-methods exploration* [Virtual symposium presentation]. Northern Colorado Human Trafficking Symposium, Fort Collins, CO.
24. Cody, A., Eng, C. L., Okech, D., Aletraris, L., & **Yi, H.** (2024, April 11–12). *Building capacity for rapid qualitative analysis in human trafficking research* [Virtual symposium presentation]. Northern Colorado Human Trafficking Symposium, Fort Collins, CO.
25. Okech, D., **Yi, H.**, Goulart, P., Waswa, A., Aletraris, L., & Clay-Warner, J. (2023, November 12–15). *Vulnerability of children with disabilities in trafficking: A low-resourced country study* [Poster]. American Public Health Association, Atlanta, GA.
26. Datta, S. G., Li, J., **Yi, H.**, Clay-Warner, J., & Okech, D. (2023, August 7–11). *Small area prevalence estimation of child trafficking at lower administrative levels of a country.* Joint Statistical Meetings, Toronto, ON, Canada. (Accepted but not presented)
27. Okech, D., **Yi, H.**, Clay-Warner, J., & Aletraris, L. (2023, January 14). *Prevalence estimation for hard-to-reach populations: Using the network scale-up method for estimating*

child trafficking prevalence in a low-resourced country [Workshop]. Society for Social Work and Research, Phoenix, AZ.

28. **Yi, H.**, Vincent, K., Okech, D., Clay-Warner, J., Li, J., Kawashima, T., & Edgemon, T. (2022, August 6). *Comparison of direct estimation and network scale-up method for prevalence estimation of child trafficking in Sierra Leone.* Paper presented at the Joint Statistical Meetings, Washington, DC.
29. Kawashima, T., Clay-Warner, J., **Yi, H.**, Li, J., & Okech, D. (2022, May 27). *The protective effect of parents' network-based social capital on child labor exploitation in Guinea.* Paper presented at the American Sociological Association Annual Meeting, Los Angeles, CA.
30. **Yi, H.**, Okech, D., Clay-Warner, J., & Li, J. (2022, January 12). *Estimating the prevalence of child trafficking in a low-resourced country: Nesting the network scale-up method in a household survey.* Society for Social Work and Research, Washington, DC. (Accepted but not presented due to COVID-19)
31. Guan, J., **Yi, H.**, Zhang, Q., & Zhang, Z. (2021, December 17–20). *A spatial-temporal optimization model for road network maintenance and rehabilitation decision-making.* Paper presented at the 21st COTA International Conference of Transportation Professionals (CICTP), Xi'an, China.
32. Liu, E., & **Yi, H.** (2018, June). *Missingness imputation for spatial-temporal road infrastructure data.* Paper presented at the World Transportation Convention (WTC), Beijing, China.
33. **Yi, H.**, Breheny, P., Imam, N., Liu, Y., & Hoeschele, I. (2014, March). *Penalized multimarker versus single-marker regression methods for genome-wide association studies of quantitative traits.* Paper presented at the Eastern North American Region (ENAR) Spring Meeting, Baltimore, MD.

Invited Presentations

1. **Yi, H.** (2026, February 3). *Data science for good: Transforming the global response to public health, social justice, and AI education challenges.* College of Science and Technology Faculty Research and Connection Series, North Carolina A&T State University, Greensboro, NC.
2. **Yi, H.** (2025, October 28). *Combating domestic and cross-border labor trafficking in the SADC region: Advancing measurement, prevention and protection.* College of Science and Technology, North Carolina A&T State University, Greensboro, NC.
3. **Yi, H.** (2025, September 1). *Estimating prevalence in hidden populations: Advances and challenges in network scale-up, respondent-driven, and Vincent link-tracing sampling.* Professor Shu Yang's Research Group, North Carolina State University, Raleigh, NC.
4. **Yi, H.**, Okech, D., & Datta, G. (2025, June 16–20). *An application of Bayesian probit small area model towards identification of high child-trafficked districts in Sierra Leone.* AIC2025: 9th African International Conference on Statistics, Strathmore University, Nairobi, Kenya.
5. **Yi, H.** (2025, April 14). *Using statistics, data science and AI to combat the crime of trafficking in persons: Digging out the hidden evidence in data.* ARTI 6950, Institute of Artificial Intelligence, University of Georgia, Athens, GA.

6. **Yi, H.** (2025, March 26). *How can quantitative research improve the global response to human trafficking?* [Faculty Lunch and Learn Seminar]. School of Social Work, University of Georgia, Athens, GA.
7. **Yi, H.** (2025, February 28). *Prevalence estimation for probabilistic and non-probabilistic samples: Traditional survey, network scale-up, link-tracing sampling and respondent-driven sampling methods.* North Carolina A&T State University, Greensboro, NC.
8. **Yi, H.** (2024, February 22). *How can emerging statistical methodologies improve the global response to human trafficking?* [Colloquium]. Department of Statistics, University of Georgia, Athens, GA. https://www.stat.uga.edu/events/content/2024/hui-yi
9. **Yi, H.** (2024, February 15). *Estimating child trafficking prevalence in a low-resourced country: Traditional survey and network scale-up methods* [Doctoral class lecture]. SOWK 8307 Doctoral Research Practicum II, School of Social Work, University of Georgia, Athens, GA.
10. **Yi, H.**, Li, J., & Wang, T. (2023, February 24). *Network scale-up method: Challenges in low-resource contexts* [Seminar]. Future Leaders Program Seminar, Center on Human Trafficking Research & Outreach, University of Georgia, Athens, GA.
11. **Yi, H.** (2023, January 17). *Hidden in plain sight: Estimating child labor trafficking in Guinea* [Colloquium]. School of Social Work Colloquia, University of Georgia, Athens, GA.
12. **Yi, H.** (2021, June 2). *Child trafficking and child labor in Sierra Leone: A quantitative study* [Seminar]. Center on Human Trafficking Research & Outreach, School of Social Work, University of Georgia, Athens, GA.

TEACHING

North Carolina Agricultural and Technical State University (2025–present)

Fall 2026	Instructor, DASC 428 Statistical Machine Learning II & STAT 710 Statistical Deep Learning, Undergraduate & Graduate (3 credit hours)
2025–2026	Instructor, MATH 224 Introductory Statistics, Undergraduate (3 credit hours; 45–48 students) Fall 2025, Spring 2026, Summer II 2026, Fall 2026

University of Georgia (2022–2025)

2022–2024	Instructor, Future Leadership Seminar (newly created for ~10 graduate and doctoral students) Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024

Chang'an University, China (2015–2019)

2017–2018	Instructor, Advanced Statistical Analysis, Graduate (newly created; 40+ students; in English) Fall 2017, Fall 2018
2016–2019	Instructor, Mathematical Statistics, Graduate (40+ students; in Mandarin) Spring 2016, Spring 2017, Spring 2018, Spring 2019

2018	Instructor, Numerical Analysis, Graduate (40+ students; in Mandarin) Spring 2018
2015–2018	Instructor, Technical Writing for International Civil Engineering Applications in English I, Undergraduate (newly created; four classes of 40+ students) Fall 2015, Fall 2016, Fall 2017, Fall 2018
2016–2019	Instructor, Technical Writing for International Civil Engineering Applications in English II, Undergraduate (newly created; four classes of 40+ students) Spring 2016, Spring 2017, Spring 2018, Spring 2019
2016–2018	Organizer, Expert Lecture Series "International Transportation Engineering Projects: Construction, Operation, Maintenance, and Management" (newly developed; 160+ students) Fall 2016, Fall 2017, Fall 2018
2015	Instructor, CAD Engineering Drawing, Undergraduate (80+ students; in Mandarin) Spring 2015
2014–2015	Instructor, Advanced Oral English, Undergraduate (four classes of 40+ students; in English) Fall 2014, Spring 2015

STUDENT ADVISING & MENTORING

North Carolina Agricultural and Technical State University

2025–2029	Chela Wallace, Ph.D. Student, STEM Education. *Role*: Ph.D. Supervisor Interdisciplinary research advancing AI, data science, and K-12 STEM education.
2025–2029	Faustina K. Acquaah, Ph.D. Student. *Role*: Dissertation Committee Member Evaluates whether environmental exposures are associated with higher biological age measures after accounting for chronological age.
2025–2029	Kristin Johnson, Ph.D. Student. *Role*: Committee Member Examines economic development, opportunities, and the future of HBCUs.
Summer 2026	Abiodun Olalere, Ph.D. Student. *Role*: Research Intern Supervisor Machine learning, deep learning, and social network analysis.
	Accolades: Won student travel award at the 2026 Fall Technical Conference.
Spring 2026	Stephen Agyeah, M.S. Student. *Role*: M.S. Committee Member Item count and sum technique methods for prevalence estimation of sensitive phenomena.
Spring 2026	Camden Jones, M.S. Student, Data Science. *Role*: Thesis Advisor
Spring 2026	Destine (Drew) Taylor, M.S. Student, Statistics. *Role*: Thesis Committee Member
	Accolades: Won the university-level Outstanding Teaching Assistant Award.
Fall 2025	Justin Smallwood, Undergraduate Student, Statistics. *Role*: Capstone Project Advisor

University of Georgia

2023–2025	Dr. Pedro Goulart, Data Analyst. *Role*: Supervisor
2023–2025	Dr. Fahmida Afroz, Ph.D. in Social Work (May 2025). *Role*: Statistical Consultant
2021–2023	Jiacheng Li, Ph.D. Student, Statistics. *Role*: Co-Supervisor (Graduate Research Assistant) Now Quantitative Analytics Specialist, Wells Fargo.
2022–2023	Tao Wang, Ph.D. Student, Statistics. *Role*: Co-Supervisor
2021–2022	Hongju Park, Ph.D. Student, Statistics. *Role*: Co-Supervisor

Chang'an University, China

2018–2019	Jinchao Guan, Ph.D. Student. *Role*: Ph.D. Supervisor Now at Zhejiang Expressway Co., Ltd., China.
2017–2018	Enze Liu, Master's Student. *Role*: Master's Supervisor Now Postdoctoral Researcher, Hefei University of Technology, China.
2016–2018	Yang Zhao, Undergraduate and Master's Student. *Role*: Instructor and Academic Advisor Later Ph.D. student in Computer Science, Temple University.
2015–2016	Gang Liu, Undergraduate Student. *Role*: Instructor and Academic Advisor Later Research Associate, University of Wisconsin–Madison.

SERVICE TO THE PROFESSION

Editorial Service

2026–present	Associate Editor, *Statistics and Public Policy*
2025–present	Consulting Editor and Reviewer, *Teaching Educational Research Methods*

Ad-Hoc Manuscript Reviewer (2019–present)

- *Statistics and Public Policy* (IF: 0.90; 2026)
- *Journal of Statistical Computation and Simulation* (IF: 1.20; 2026)
- *Humanities and Social Sciences Communications* (IF: 3.70; 2024, 2025, 2026)
- *BMC Health Services Research* (IF: 3.40; 2026)
- *Journal of Human Trafficking* (IF: 2.05; 2024, 2026)
- *The Annals of Applied Statistics* (IF: 1.66; 2025)
- Oxford University Press (IF: 2.16; 2024)
- *Quality & Quantity* (IF: 3.43; 2024)
- *Cities* (IF: 6.08; 2023)
- *International Statistical Review* (IF: 1.95; 2019)
- *Sankhya Series B* (IF: 0.89; 2019)
- *Transportmetrica A: Transport Science* (IF: 3.28; 2019)

Conference Service

2017–2019 Technical Committee, World Transportation Convention Meetings, Beijing, China

UNIVERSITY SERVICE

North Carolina Agricultural and Technical State University (2025–present)

- Judge, University Graduate Student Research Symposium
- Volunteer, College of Science and Technology research events
- Volunteer, departmental student and faculty events

University of Georgia (2020–2025)

- Search Committee Member for assistant research scientist, data analyst, postdoc, and Ph.D. graduate assistant positions

Chang'an University, China (2015–2019)

- Undergraduate Recruitment Committee
- Undergraduate Curriculum Committee
- Undergraduate Academic Advisor
- Professional Master Program Committee
- Ph.D. Program Committee
- International Talents Hiring Committee
- Office of Institutional Research, Program Review and Assessment Committee
- Office of Provost, Task Force on Academic Excellence

PROFESSIONAL AFFILIATIONS

2026	Faculty Cohort, Amazon Web Services Machine Learning University
2022–present	American Statistical Association Sections: Statistical Computing; Statistical Learning and Data Science; Survey Research Methods; Social Statistics; Statistics in Epidemiology
2020–present	Society for Social Work and Research Sections: Research Design and Measurement; Violence against Women and Children; International Social Work & Global Issues; Communities and Neighborhoods
2022–2023	Georgia Statewide Human Trafficking Task Force

MEDIA COVERAGE

1. Gallant, A. (2024, March 5). UGA receives $5.2M to study labor trafficking. *UGA Today*. https://news.uga.edu/uga-receives-5-2m-to-combat-labor-trafficking/
2. Randal, O. (2024, April 2). UGA to recognize groundbreaking achievements at 2024 Research Awards. *UGA Research*. https://research.uga.edu/news/uga-to-recognize-groundbreaking-achievements-at-2024-research-awards/

3. Aletraris, L., Cody, A., & **Yi, H.** (2024). 2023 Impact Report. Center on Human Trafficking Research & Outreach, University of Georgia. https://cenhtro.uga.edu/about/annual-impact-reports/2023-impact-report/research-innovation/

4. **Yi, H.**, & colleagues. (2023). *Severe forms of sex trafficking in the gold mining areas of Kédougou, Senegal*. Featured research report, Human Trafficking Search. https://humantraffickingsearch.org/resource/severe-forms-of-sex-trafficking-in-the-gold-mining-areas-of-kedougou-senegal/

5. Gallant, A. (2022, October 7). UGA receives $2.2M to study labor trafficking. *UGA Today*. https://news.uga.edu/uga-receives-2-2m-to-study-labor-trafficking/